Vitamins, Minerals & Water – Page 423

Source: Lucent's General Knowledge – Biology

Biology: Vitamins, Minerals, Water & Balanced Diet

50 Questions • 30 Minutes

Instructions:
- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. What is the chemical name of Vitamin D? **A.** Tocopherol **B.** Calciferol **C.** Phylloquinone **D.** Retinol	**Ans: B**
Q2. Deficiency of Vitamin D in children causes: **A.** Scurvy **B.** Beriberi **C.** Rickets **D.** Night blindness	**Ans: C**
Q3. What is the chemical name of Vitamin E? **A.** Calciferol **B.** Ascorbic acid **C.** Tocopherol **D.** Thiamine	**Ans: C**
Q4. Deficiency of Vitamin E causes: **A.** Non-clotting of blood **B.** Less fertility and destruction of RBC **C.** Night blindness **D.** Rickets	**Ans: B**
Q5. Sources of Vitamin E include: **A.** Tomato and Soybean oil **B.** Leafy vegetables, Milk, Butter, Sprouted wheat, Vegetable oil **C.** Sea fish and Iodized salt **D.** Meat and Liver only	**Ans: B**
Q6. What is the chemical name of Vitamin K? **A.** Phylloquinone **B.** Riboflavin **C.** Niacin **D.** Pyridoxine	**Ans: A**
Q7. Deficiency of Vitamin K causes: **A.** Scurvy **B.** Pellagra **C.** Non-clotting of blood **D.** Anemia	**Ans: C**

Q8. Cobalt is found in which vitamin? **A.** Vitamin A **B.** Vitamin C **C.** Vitamin B12 **D.** Vitamin D	**Ans: C**
Q9. Synthesis of Vitamin D takes place through which substance in sunlight? **A.** Keratin **B.** Melanin **C.** Cholesterol (Ergosterol) **D.** Hemoglobin	**Ans: C**
Q10. Vitamin D2 is also known as: **A.** Calciferol **B.** Ergocalciferol **C.** Tocopherol **D.** Phylloquinone	**Ans: B**
Q11. Vitamin K is synthesized in our body by: **A.** Liver cells **B.** Kidney cells **C.** Bacteria in the colon **D.** Skin cells	**Ans: C**
Q12. Which vitamin is mainly stored in the liver? **A.** Vitamin C **B.** Vitamin B1 **C.** Vitamin A **D.** Vitamin E	**Ans: C**
Q13. Use of polished rice in human diet causes: **A.** Scurvy **B.** Rickets **C.** Beriberi **D.** Pellagra	**Ans: C**
Q14. Which is the richest source of Vitamin C? **A.** Milk **B.** Egg **C.** Amla **D.** Fish	**Ans: C**

Q15. Bleeding gum, falling of teeth, fragile bones, and delayed wound healing are symptoms of deficiency of: **A.** Vitamin A **B.** Vitamin D **C.** Vitamin C **D.** Vitamin K	**Ans: C**

Q16. Deficiency of calcium mainly occurs in the absence of which vitamin? **A.** Vitamin A **B.** Vitamin B **C.** Vitamin C **D.** Vitamin D	**Ans: D**

Q17. The daily requirement of Sodium (as sodium chloride) is: **A.** 20 mg **B.** 1 gram **C.** 2–5 grams **D.** 1.2 grams	**Ans: C**

Q18. Sodium is normally found in which part of the cell? **A.** Nucleus **B.** Protoplasm **C.** External fluid of cell **D.** Mitochondria	**Ans: C**

Q19. The daily requirement of Potassium is: **A.** 2–5 grams **B.** 1 gram **C.** 20 mg **D.** 1.2 grams	**Ans: B**

Q20. Potassium is normally found in: **A.** External fluid of cell **B.** Protoplasm **C.** Bones only **D.** Blood plasma only	**Ans: B**

Q21. The daily requirement of Calcium is approximately: **A.** 20 mg **B.** 1 gram **C.** 1.2 grams **D.** 2–5 grams	**Ans: C**

Q22. **Which mineral provides strength to bones and teeth, and helps in blood clotting?**

A. Iron
B. Calcium
C. Zinc
D. Iodine

Ans: B

Q23. **The daily requirement of Iron for boys and girls respectively is:**

A. 20 mg for both
B. 25 mg (boy) and 35 mg (girl)
C. 35 mg (boy) and 25 mg (girl)
D. 10 mg for both

Ans: B

Q24. **Iron is an important component of:**

A. Thyroxin hormone
B. Insulin
C. Red Blood Corpuscles and Haemoglobin
D. Bones and Teeth

Ans: C

Q25. **Iodine is important for the synthesis of which hormone?**

A. Insulin
B. Adrenaline
C. Thyroxin hormone secreted by Thyroid gland
D. Growth hormone

Ans: C

Q26. **The daily requirement of Iodine is:**

A. 1 gram
B. 1.2 grams
C. 20 mg
D. 2–5 grams

Ans: C

Q27. **Which mineral is required for insulin functioning?**

A. Copper
B. Magnesium
C. Zinc
D. Cobalt

Ans: C

Q28. **Magnesium is important for functioning of:**

A. Thyroid gland
B. Muscular system and nervous system
C. Insulin production
D. Blood clotting

Ans: B

Q29. **Copper is needed for:**

A. Insulin functioning
B. Thyroxin synthesis
C. Formation of haemoglobin and bones, and as a conductor of electron
D. Nerve impulse transmission only

Ans: C

Q30. Cobalt is essential for synthesis of: **A.** Thyroxin and Growth hormone **B.** RBC and Vitamin B12 **C.** Insulin and Glucagon **D.** Hemoglobin only	**Ans: B**

Q31. Deficiency of fluoride leads to: **A.** Anemia **B.** Goitre **C.** Mottling of teeth **D.** Night blindness	**Ans: C**

Q32. Presence of excess fluorine in water causes: **A.** Goitre **B.** Fluorosis **C.** Rickets **D.** Scurvy	**Ans: B**

Q33. To reduce tooth decay, most toothpastes contain: **A.** Calcium **B.** Iron **C.** Fluoride **D.** Zinc	**Ans: C**

Q34. The hard enamel layer of teeth is made of: **A.** Calcium carbonate **B.** Calcium hydroxy apatite **C.** Calcium oxalate **D.** Calcium phosphate only	**Ans: B**

Q35. Deficiency of calcium and iron is generally found in: **A.** Children **B.** Elderly men **C.** Pregnant women **D.** Athletes	**Ans: C**

Q36. A typical adult human body contains about how much magnesium? **A.** 5 grams **B.** 10 grams **C.** 25 grams **D.** 50 grams	**Ans: C**

Q37. Water is what percentage of our body weight? **A.** 40% **B.** 50% **C.** 60% **D.** 70%	**Ans: C**

Q38. Which of the following is NOT a main function of water in the body? **A.** Controls temperature by sweating and vapourization **B.** Important for excretion of excretory substances **C.** Synthesis of Vitamin D **D.** Maximum chemical reactions through hydrolysis	**Ans: C**

Q39. Minerals are: **A.** Organic compounds **B.** Homogenous inorganic material needed for metabolism **C.** Proteins **D.** Carbohydrates	**Ans: B**

Q40. Sources of Iodine include: **A.** Milk and cheese only **B.** Sea fish, sea food, green leafy vegetables, iodized salt **C.** Meat and liver only **D.** Eggs and bread only	**Ans: B**

Q41. Which mineral deficiency affects the activity of nitrogenase, nitrate reductase, and chlorate reductase in plants? **A.** Zinc **B.** Copper **C.** Molybdenum **D.** Magnesium	**Ans: C**

Q42. Sodium is related to which function in the body? **A.** Insulin functioning **B.** Contraction of muscles and transmission of nerve impulses **C.** Synthesis of thyroxin **D.** Formation of haemoglobin	**Ans: B**

Q43. Main sources of Calcium include: **A.** Sea fish and iodized salt **B.** Liver and fishes **C.** Milk, cheese, eggs, grains, fish **D.** Meat, fish, liver and grains	**Ans: C**

Q44. Phosphorus daily requirement is: **A.** 20 mg **B.** 1.2 grams **C.** 1 gram **D.** 25 mg	**Ans: B**

Q45. Phosphorus provides strength to bones and teeth in coordination with: **A.** Iron **B.** Zinc **C.** Calcium **D.** Iodine	**Ans: C**

Q46. Synthesis of vitamins can/cannot be done by cells of the body: **A.** Can be done easily **B.** Cannot be done; fulfilled by vitamin-containing food **C.** Only in liver **D.** Only in kidneys	**Ans: B**

Q47. Which vitamins can be synthesized in the human body? **A.** Vitamin A and B **B.** Vitamin B and C **C.** Vitamin D and K **D.** Vitamin E and K	**Ans: C**

Q48. Sources of Vitamin K include: **A.** Fish liver oil and milk **B.** Tomato, Soybean oil, Green vegetables, Alfalfa **C.** Eggs and bread **D.** Meat and liver	**Ans: B**

Q49. Sources of Iron include: **A.** Sea fish and iodized salt **B.** Vegetables only **C.** Albumen of egg, bread, Bajra, Banana, Spinach, Apple **D.** Milk and cheese only	**Ans: C**

Q50. Balance Diet is defined as: **A.** Eating only proteins **B.** Nutrition in which all important nutrients are available in sufficient quantity **C.** Eating only carbohydrates **D.** Fasting for health	**Ans: B**

Answer Key & Explanations

Q1. Answer: B — Calciferol
Page 423: Vitamin D chemical name is Calciferol.

Topic: Vitamins

Q2. Answer: C — Rickets
Page 423: Vitamin D deficiency causes Rickets in children and Osteomalacia/Osteoporosis in adults.

Topic: Vitamins

Q3. Answer: C — Tocopherol
Page 423: Vitamin E chemical name is Tocopherol.

Topic: Vitamins

Q4. Answer: B — Less fertility and destruction of RBC
Page 423: Vitamin E deficiency causes less fertility and destruction of RBC.

Topic: Vitamins

Q5. Answer: B — Leafy vegetables, Milk, Butter, Sprouted wheat, Vegetable oil
Page 423: Vitamin E sources include Leafy vegetables, Milk, Butter, Sprouted wheat, Vegetable oil, Wheat germ oil.

Topic: Vitamins

Q6. Answer: A — Phylloquinone
Page 423: Vitamin K chemical name is Phylloquinone.

Topic: Vitamins

Q7. Answer: C — Non-clotting of blood
Page 423: Vitamin K deficiency causes non-clotting of blood.

Topic: Vitamins

Q8. Answer: C — Vitamin B12
Page 423: Cobalt is found in Vitamin B12.

Topic: Vitamins

Q9. Answer: C — Cholesterol (Ergosterol)
Page 423: Synthesis of Vitamin D takes place through cholesterol (ergosterol) present in skin via ultraviolet rays of sunlight.

Topic: Vitamins

Q10. Answer: B — Ergocalciferol
Page 423: Vitamin D2 is known as ergocalciferol.

Topic: Vitamins

Q11. Answer: C — Bacteria in the colon
Page 423: Vitamin K is synthesized in our colon by bacteria and from there it is absorbed.

Topic: Vitamins

Q12. Answer: C — Vitamin A

Page 423: Vitamin A is mainly stored in liver.

Topic: Vitamins

Q13. Answer: C — Beriberi

Page 423: Use of polished rice in human diet causes Beriberi.

Topic: Vitamins

Q14. Answer: C — Amla

Page 423: Amla is the richest source of Vitamin C.

Topic: Vitamins

Q15. Answer: C — Vitamin C

Page 423: These symptoms occur due to deficiency of Vitamin C.

Topic: Vitamins

Q16. Answer: D — Vitamin D

Page 423: Deficiency of calcium mainly occurs in absence of Vitamin D.

Topic: Vitamins

Q17. Answer: C — 2–5 grams

Page 423: Daily requirement of Sodium is 2–5 grams (as sodium chloride).

Topic: Minerals

Q18. Answer: C — External fluid of cell

Page 423: Sodium is normally found in external fluid of cell.

Topic: Minerals

Q19. Answer: B — 1 gram

Page 423: Daily requirement of Potassium is approximately 1 gram.

Topic: Minerals

Q20. Answer: B — Protoplasm

Page 423: Potassium is normally found in protoplasm. It is important for different chemical reactions in cells.

Topic: Minerals

Q21. Answer: C — 1.2 grams

Page 423: Daily requirement of Calcium is approximately 1.2 grams.

Topic: Minerals

Q22. Answer: B — Calcium

Page 423: Calcium provides strength to bones and teeth with vitamin, important role in blood formation, muscular contraction, and help in clotting of blood.

Topic: Minerals

Q23. Answer: B — 25 mg (boy) and 35 mg (girl)
Page 423: Iron daily requirement is 25 mg for boys and 35 mg for girls.

Topic: Minerals

Q24. Answer: C — Red Blood Corpuscles and Haemoglobin
Page 423: Iron is important component of Red Blood Corpuscles and haemoglobin.

Topic: Minerals

Q25. Answer: C — Thyroxin hormone secreted by Thyroid gland
Page 423: Iodine is important for synthesis of thyroxin hormone secreted by Thyroid gland.

Topic: Minerals

Q26. Answer: C — 20 mg
Page 423: Daily requirement of Iodine is 20 mg.

Topic: Minerals

Q27. Answer: C — Zinc
Page 423: Zinc is required for insulin functioning. Sources include liver and fishes.

Topic: Minerals

Q28. Answer: B — Muscular system and nervous system
Page 423: Magnesium is for functioning of muscular system and nervous system.

Topic: Minerals

Q29. Answer: C — Formation of haemoglobin and bones, and as a conductor of electron
Page 423: Copper is needed for formation of haemoglobin and bones and as a conductor of electron.

Topic: Minerals

Q30. Answer: B — RBC and Vitamin B12
Page 423: Cobalt is essential for synthesis of RBC and Vitamin B12.

Topic: Minerals

Q31. Answer: C — Mottling of teeth
Page 423: Deficiency of fluoride leads to mottling of teeth.

Topic: Minerals

Q32. Answer: B — Fluorosis
Page 423: Presence of excess fluorine in water causes fluorosis.

Topic: Minerals

Q33. Answer: C — Fluoride
Page 423: To reduce tooth decay most tooth pastes contain fluoride.

Topic: Minerals

Q34. Answer: B — Calcium hydroxy apatite
Page 423: Hard enamel layer of teeth is calcium hydroxy apatite.

Topic: Minerals

Q35. Answer: C — Pregnant women
Page 423: Deficiency of calcium and iron is generally found in pregnant women.

Topic: Minerals

Q36. Answer: C — 25 grams
Page 423: A typical adult human body contains about 25 grams of magnesium.

Topic: Minerals

Q37. Answer: C — 60%
Page 423: Water is the important component of our body, 60% weight of the body is water.

Topic: Water

Q38. Answer: C — Synthesis of Vitamin D
Page 423: Water functions include temperature control, excretion, and hydrolysis. Vitamin D synthesis is by sunlight/cholesterol, not water.

Topic: Water

Q39. Answer: B — Homogenous inorganic material needed for metabolism
Page 423: Mineral is a homogenous inorganic material needed for body which control the metabolism of body.

Topic: Minerals

Q40. Answer: B — Sea fish, sea food, green leafy vegetables, iodized salt
Page 423: Iodine sources include sea fish, sea food, green leafy vegetables, iodized salt.

Topic: Minerals

Q41. Answer: C — Molybdenum
Page 423: Deficiency of molybdenum affects the activity of nitrogenase, nitrate reductase and chlorate reductase in plants.

Topic: Minerals

Q42. Answer: B — Contraction of muscles and transmission of nerve impulses
Page 423: Sodium is related to contraction of muscles and transmission of nerve impulses in nerve fiber.

Topic: Minerals

Q43. Answer: C — Milk, cheese, eggs, grains, fish
Page 423: Calcium sources include milk, cheese, eggs, grains, gram, fish etc.

Topic: Minerals

Q44. Answer: B — 1.2 grams
Page 423: Phosphorus daily requirement is 1.2 gram. Sources include milk, cheese, bajra, green leafy vegetables.

Topic: Minerals

Q45. Answer: C — Calcium

Page 423: Phosphorus provides strength to bones and teeth, in coordination with calcium.

Topic: Minerals

Q46. Answer: B — Cannot be done; fulfilled by vitamin-containing food

Page 423: Synthesis of vitamins cannot be done by the cells of body. It is fulfilled by vitamin containing food.

Topic: Vitamins

Q47. Answer: C — Vitamin D and K

Page 423: Synthesis of Vitamin D and K takes place in body (D via sunlight, K via colon bacteria).

Topic: Vitamins

Q48. Answer: B — Tomato, Soybean oil, Green vegetables, Alfalfa

Page 423: Vitamin K sources include Tomato, Soybean oil, Green vegetables, Alfalfa etc.

Topic: Vitamins

Q49. Answer: C — Albumen of egg, bread, Bajra, Banana, Spinach, Apple

Page 423: Iron sources include albumen of egg, bread, Bajra, Banana, Spinach apple.

Topic: Minerals

Q50. Answer: B — Nutrition in which all important nutrients are available in sufficient quantity

Page 423: Balance Diet is that nutrition in which all the important nutrients for organism are available in sufficient quantity.

Topic: Balanced Diet